Name :- Laiba Fatima

Class :- BS AI

Semester :- 5th

Section :- A

Reg # :- 2023-BS-AI-047

After Mids

" Probability "

$$0 \le P \le 1$$

impossible event — sure event

$$P(A) = \frac{n(A)}{n(S)}$$

P(A) → any event

n(S) → sum of numbers

$$\text{Prop} = \frac{\text{fav. element}}{\text{Total Possible element}}$$

Experiment: Any planned activity or any process which generates data. A single performance of an experiment is called "trial".

Random Experiment: Any planned activity or any process which generates different data under similar condition.

Sample Pace: Denoted by (S), A complete set of all possible elements from an experiment is called sample space.

Element: An outcome of experiment is called element. Each outcome in sample space.

Event: A complete set of all favourable outcomes from sample space. Denoted by [A,B,C] capital letters is called event. Event should be subset of sample space.

Wed, 22nd Oct. 2025

Types of Events

1- Simple Event :-

An event consisting only one element is called simple event.

Example:- $S = \{1,2,3,4,5,6\} \rightarrow A = \{\text{multiple of } 5\}$

$S = \{5\}$ $A = 5 \rightarrow$ simple

2- Compound event :-

An event consisting of more than one element.

Example:- $S = \{\text{even no.}\}$

$B = \{2,4,6\} \rightarrow$ compound event

Tuesday, 4th Nov. 2025

"Types of events"

1- Equally likely events:-

When probability on one event is equal to the probability of second event.

2- Mutually exclusive measure:-

imp

Events that cannot occur together / same time. def + apply

$P(A \cap B) = 0$ → no common probability

Q:- In a company 60 percent of the trainees will remain with a company. In the same company, 50 percent of the employees earn more than 10,000 per month. Is this mutually exclusive

$P(T) = 0.60$

$P(E) = 0.50$

yes, it can be

Independant event:

When the occurence of one event does not affect the occurence of another event.

$P(A \cap B) = P(A) \cdot P(B)$

ADDITION LAW,

$P(A \text{ or } B) \Leftrightarrow P(A \cup B)$

$P(A) + P(B)$

When events are mutually exclusive

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Q: In a company 60 percent of the trainees will remain with a company. In the same company, 50 percent of the employees earn more than 10,000 per month.
Find the probability that an employee who is a trainee or the employee earn more than 10,000.

Given that the probability of internee and employee who earn more than 10000 is 40 percen

$0.60 + 0.5 - [0.4]$
$= 1.1 - 0.7$
$= 70\%$

probability can never be greater than 1.

Q: A dice is rolled twice, what is the probability that 1st toss shows an even number and the 2nd toss shows a 6.

$P(\text{even} \cap \text{six})$

→ now check for events are dependent or independent

$= P(\text{even}) \cdot P(\text{six})$
$= \frac{1}{2} \cdot \frac{1}{6}$
$= \frac{1}{12}$

- $P(A \cup B) = P(A) + P(B)$ → when mutually exclusive
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

When Not | Multiplication Law

U → or → check events

→ when dependent multiplication nhi hogi

Wednesday, 12th Nov. 2025

Dependant vs Independant events

Addition Law vs Multiplication Law

Conditional Probability

∴ intersection always from total

Multiplication Law:-

$$P(A \cap B) = P(A) \cdot P(B)$$

if both independant

$$P(A \cap B) = P(A) \cdot P(B/A)$$
$$= P(B) \cdot P(A/B)$$

if dependant

Q:- A professor thinks students who live on campus are more likely to get an A grade in probability course, to check this theory Hamza rizwan collects data from past few years and he found that 600 student have taken this course. 120 students got A grade, 200 students live on campus, 80 of the students live off campus and got A grade. Q1 If a student is selected randomly, what is the probability that he gets A grade. Q2 What is the probability that a student lives off campus. Q3 Find the probability that a student live on campus or do not gets A grade.

Q4 Find the probability that a student did not get A grade and he lives on campus. If selected student gets A grade what is the probability that he or she lives off campus? 96 Find the probability that a student lives on campus or he lives at home.

	ON	OFF	
A	40	80	120
A'	160	320	480
	200	400	600

1- $P(A) = \frac{120}{600}$

2- $P(\text{off}) = \frac{400}{600}$

3- $P(\text{on} \cup A') = P(\text{on}) + P(A') - P(\text{on} \cap A')$

$= \frac{200}{600} + \frac{480}{600} - \frac{160}{600}$

$=$

4- $P(A' \cap \text{ON}) = \frac{160}{600}$

5- $P(\text{off} / A) = \frac{P(\text{off} \cap A)}{P(A)}$

$\frac{80}{120} = \frac{80/600}{120/60}$

6- $\frac{600}{600} = 1 \rightarrow 100\%$

Q:- If two dices are rolled together and same number occurs on each dice, what is the probability that first digit is even.

$= \frac{3}{6}$

$= \frac{1}{2}$

Sample → space → even

Conditional probability
↳ Cross tab

Baye's Theorem
↳ Excluded

Binomial Probability
distribution

Tuesday, Nov. 18-2025

Worksheet Class Assignment

1. $P(cat) = 0.4$

$P(correct \mid cat) = 0.95$

find : $P(correct \text{ AND } cat)$

$P = P(cat) \times P(correct \mid cat) = 0.4 \times 0.95 = 0.38$

$\boxed{\text{Answer : } 0.38}$

2. 60% servers active $\rightarrow P(A) = 0.6$
40% of active servers $\rightarrow P(U \mid A) = 0.4$
25% of all servers have update $\rightarrow P(U) = 0.25$

$P(A \cap U) = P(A) \times P(U \mid A) = 0.6 \times 0.4 = 0.24$

$P(A \mid U) = \dfrac{P(A \cap U)}{P(U)} = \dfrac{0.24}{0.25} = 0.96$

3. $P(G) = 0.30$
$P(L) = 0.50$
$P(G \cap L) = 0.20$

a) $P(G \cup L) = P(G) + P(L) - P(G \cap L)$
$= 0.30 + 0.5 - 0.20$
$= 0.8 - 0.20 = 0.6$

(b) - ~~Yes~~ No, they are not mutually exclusive

4- spam = 20%
actually contain = 15%

probability of both containing malicious and mark as spam

$0.2 \times 0.15 = 0.03$

5 (a)- $P(A) = 0.8$
$P(B) = 0.7$

$P(A \cap B) = 0.6$

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.7 - 0.6$
$= 0.9$

(b)- No, not independant

$P(A \cap B) = P(A) \times P(B) = 0.8 \times 0.7 = 0.56 \neq 0.6$

6- $P(A) = 0.75$
$P(F) = 0.60$
$P(A \cap F) = 0.45$

(a)- Probability $= P(A) + P(F) - P(A \cap F)$
$= 0.75 + 0.60 - 0.45$
$= 0.9$

$P(A \cap F) = P(A) \times P(F) = 0.75 \times 0.6 = 0.45$

7- Yes independant

$P(A) = 40\% = 0.4$
$P(B) = 35\% = 0.35$
$P(Both) = 20\% = 0.2 \leftarrow P(A \cap B)$

(a) probabil. $= 0.4 + 0.35 - 0.2$
$= 0.55$

(b) $0.35 \times 0.4 = .14 \neq 0.2$ so not independent

8- p(overload) = 0.10
p(crash) = 0.05
p(O ∩ C) = 0.01

They are not mutually exclusive

prob = 0.10 + 0.05 - 0.01
= 0.14

0.10 × 0.05 = not independent
↳ its dependent

9-

26th Nov. 2025 / Wednesday

"Probability distribution"

Discrete Probability Distribution

Continuous Probability Distribution

* Binomial Probability Distribution:-

• The outcome is divided into two categories. Yes, No, Right, Wrong etc.
• Trials are independant.
• Probability of success remains same throughout the trials.
• Trials are fixed.

$\boxed{\therefore n \text{ is fixed}}$

$$P(X = x) = \binom{n}{x} \cdot p^{x} \cdot q^{n-x}$$

$x = 0, 1, 2 \ldots\ldots n$

$\boxed{p+q = 1}$

→ p = prob of success
→ q = prob of failure
→ n = no. of trials / samples

Mean, Binonimal ka mean

$p \times n$

Parameters:-

p and n are the parameters

Mean and variance:-

Mean $= n \cdot p$
var $= n \cdot p \cdot q$
SD $= \sqrt{n \cdot p \cdot q}$

~~Q:- If we make a code for Data Analysis The probability that the code execute properly and we~~

Q:- If we make a code for error detection, the probability that this code detects errors, 70 % of the time, if we run this code 6 times, what is the probability that..

(a)- This code detects exactly all the ~~errors~~ times /in each trial.

(b)- detects exactly 2 times

(c)- at least one time

(d)- at most 4 times

(e)- between 3 to 5 times

(f)- code does not detect error 2 times.

Answers:-

$q = 0.30$

$p = 0.70$

$n = 6$

$X = 0, 1, 2, 3, 4, 5, 6$

1 $P(X = 6) = ?$

2 - $P(X = 2)$

3 - $P(x \geq 1) = P(1) + P(2) + \dots P(6)$

4 - $P(X \leq 4) = P(0) + P(1) + P(2) + P(3) + P(4)$

~~5 - $P(X \leq 4) = P(0) + P(1$~~

5 - $P(3 < x < 5) = P(X = 4)$

6 - $P(x = 2)$

Not solving yet, only looking for probability:

4 decimal places

$$P(x=0) = \binom{6}{0} (0.70)^0 (0.30)^{6-0} = 0.0007$$

$$P(x=1) = \binom{6}{1} (0.70)^1 (0.30)^{6-1} = 0.010$$

$$P(x=2) = \binom{6}{2} (0.70)^2 (0.30)^{6-2} = 0.059$$

$$P(x=3) = \binom{6}{3} (0.70)^3 (0.30)^{6-3} = 0.185$$

$$P(x=4) = \binom{6}{4} (0.70)^4 (0.30)^{6-4} = 0.324$$

$$P(x=5) = \binom{6}{5} (0.70)^5 (0.30)^{6-5} = 0.3025$$

$$P(x=6) = \binom{6}{6} (0.70)^6 (0.30)^{6-6} = 0.1176$$

X	P(X)
0	0.0007
1	0.010
2	0.059
3	0.185
4	0.324
5	0.3025
6	0.1176
+	1

Now Answers

1- $P(x=6) = 0.1176$

2- $P(x=2) = 0.059$

3- $P(x \geq 1) = 1 - P(0)$

$= 1 - 0.0007$

$= 0.9993$

4- $P(x \leq 4) = 1 - [P(5) + P(6)]$

$= 1 - [0.3025 + 0.1176]$

$= 0.5799$

5- $P(3 < x < 5) = P(x = 4) = 0.324$

6- $P(x = 2) = X$ = Not detect

Now $p = 0.30$, $q = 0.70$

$P(X = 2) = \binom{6}{2} (0.30)^2 (0.70)^4$

=

Q:- A company sends promotional emails to their customers, 30% of the customers usually responds (i)- if the company send 12 emails. What is the probability that exactly 5 customers will respond (ii)- at least 2 customers will respond (iii)- at least 1 will not respond

If the company sends 12 emails, how many customers will respond and how many will not respond.

$p(A) = 0.3$ $\qquad$ $p(X) = 0.7$

(i)- $p(x = 5) = \binom{12}{5} (0.30)^5 (0.70)^7 =$

(ii) $P(P \geq 2) = 1 - [P(0) + P(1)]$

$= 1 - [\qquad]$

(iii) - $P(X \geq 1)$ - at least one will not respond,

Now $p = 0.70$, $q = 0.30$

$= 1 - [P(0)]$ $\qquad p(x=0) = \binom{12}{0}(0.70)^0(0.30)^{12}$

$= 1 - [\quad]$ $\qquad = 1 - [\quad]$

$$p(x=0) = \binom{12}{0}(0.30)^0(0.70)^{12}$$

$$p(x=1) = \binom{12}{1}(0.30)^1(0.70)^{11}$$

(iv) - They will respond	They will not respond
0.30, $n = 12$	$n = 12$
Mean $= 12 \times 0.30$	12×0.70
$= 3.6$	≈ 8.4
≈ 4	8
because 3.6 humans can't be	
so 4	

Q:- In a factory 30 out of 70 bulbs are defective, HR team will test and check, 7 bulbs. What is the probability that all the bulbs are defective.

Not defective $\frac{40}{70} \times 100 = 57\%$. 0.57

defective $\frac{30}{70} \times 100 = 42.8\%$ 0.42

$n = 7$

$P = \frac{X}{N}$

$= \frac{30}{70}$

$P(X = 7) = \binom{7}{7} (0.42)^7 (0.58)^0$

$=$

3rd Dec. 2025 / Wednesday

"Poisson Probability Distribution"

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

λ = parameter (Mean, average)

$x = 0, 1, 2 \ldots \infty$

→ Mean and Var

don't roundoff

Mean = its parameter (λ) = Var

Q:- On average 6 students enter in the room (i). in 60 min (i). What is the probability 3 ~~all~~ the students enter in the room.
(ii). At least 1 student enter in the room in 30 min.

Answer: (i) $P(x=3) = \frac{e^{-6} \cdot 6^3}{3!} = 0.089$ $\lambda = 6$ per hour

(ii). $P(x \geq 1) = 1 - [P(0)]$ $\lambda = 0.1$ per min

$= 1 - \left[\frac{e^{-3} \cdot 3^0}{0!}\right]$ $\lambda = 0.1 \times 30$

$= 0.951$ $\lambda = 3$ per 30 min

→ depends on λ lamda whether to roundoff or not
if data is discrete round off number of outputs point me ni hogi

Q:- A program execution time is X   time is continuous

Q:- A specefic program detects 4 errors in 10 min on average. If we execute the program what is the probability that the program detects error between 5 to 9 errors in 15 min.

~~6+7+8 = 21~~

$\lambda = 4$ error in 10 min

$\lambda = 0.4$ / per min

~~P (x =~~

in 15 min

$\lambda = 6$

$P(5 < x < 9)$

$= P(6) + P(7) + P(8)$

$\lambda = 6$

$P(x=6) = \frac{e^{-6}\, 6^{6}}{6!}$

$P(x=7) = \frac{e^{-6}\, 6^{7}}{7!}$

$P = (x=8) = \frac{e^{-6}\, 6^{8}}{8!}$

$P(x=6) + P(x=7) + P(x=8)$

$\frac{e^{-6}\, 6^{6}}{6!} + \frac{e^{-6}\, 6^{7}}{7!} + \frac{e^{-6}\, 6^{8}}{8!}$

9th Dec. 2025 / Tuesday

"Regression and correlation"

Regression is a process by which we estimate or predict the value of dependant variable on the basis of one or more independant variable. If there is one independant and 1 dependent variable then the regression is called simple linear regression.

→ If there are more than 1 independant variable and 1 dependant variable in the model, then this regression is called multiple linear regression.

+ Types (Simple + Multiple)

Correlation : measure Strength of Association, Relation

How to check accuracy of regression model.

-1 to 1 ← values lie between

$Y = a + bx$ → Regression eq / Prediction model

Y → dependant

x → independant

a, b → coefficient of Reg model

$a = \bar{y} - b\bar{x}$

mean of dep → $\bar{y}$

mean of indep variable → $\bar{x}$

$$b = \frac{n \cdot \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$\sum xy$ → product ka sum

$(\sum x)(\sum y)$ → sum ka product

a = intercept

b = slope

slope: if one unit change in x (independent) variable, the value of dependent variable, will be increased or decreased by B.(increased if slope is positive and decreased if slope is negative.)

X	Y	xy	x^2
$ adv cost	sales $		
3	7	21	9
5	10	50	25
4	9	36	16
2	7	14	4
+ 1	+ 5	+ 5	+ 1
15	38	126	55

* Make a prediction model
* Predict the sales if we insist 9$ on adv cost
* make a reg line / eq

$a = \bar{y} - b\bar{x}$ | $b = \frac{n\Sigma xy - \Sigma x\Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$

Regression model

↳ $\hat{y} = a + bx$

prediction value

$b = \frac{5(126) - (15)(38)}{5(55) - (15)^2}$

$= 1.2$

$n = 5$

$\Sigma xy = 126$

$\Sigma x^2 = 55$

$\Sigma x = 15$

$\Sigma y = 38$

$a = 7.6 - 1.2(3) = 4$

$\bar{x} = \frac{\Sigma x}{n} = \frac{15}{5} = 3$

$\bar{Y} = \frac{\Sigma y}{n} = \frac{38}{5} = 7.6$

Regression model :- $\hat{y} = a + bx$

$\hat{sales} = a + b\,(\text{adv cost})$

$\hat{sales} = 4 + 1.2\,(\text{ad cost})$

If one unit change in advertisement cost the sales will increase with 1.2, if advertisement cost = 0, then average sales would be equal to 4.

⇒ if advertisement cost would be equal to 9 dollars, our predicted sales will be

$\hat{sales} = 4 + 1.2\,(\text{adv cost})$

$= 4 + 1.2(9)$

$\hat{sales} = 14.8$

23rd Dec·2025, Tuesday

"Regression and correlation"

$\hat{y} = a + bx$

a, b → coefficient of reg model

$a = \bar{y} - b\bar{x}$

$$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

a = intercept

b = slope

→ it define the change in dependant due to independant (also b define the relationship).

Correlation :-

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{[n\sum x^2 - (\sum x)^2][n \cdot \sum y^2 - (\sum y^2)]}$$

always falls b/w -1 to +1

$-1 \leq r \leq +1$

Coefficient of determination (R^2)

$R^2 = (r)^2$

Means for finding R^2 you just have to take sq of correlation R^2 always positive and falls between 0 to 1.

Interpretation → lazmi krni

If mid, estimate F.S is

$\hat{FS} = -0.12 + 1.41(15)$
$\hat{FS} = 21.03$

Example.

Here unit is marks
If one unit increase, the final score will be increased by 1.4

Interpretation: if one mark is increased in Mid Score the FS will also be increased by 1.41.

X	Y	xy	x^2	y^2
3	7	21	9	for correlation
2	1	2	4	
8	10	80	64	
5	9	45	25	
3	2	6	9	

Correlation: .5 ↓ weak   .5 ↗ strong

∴ coefficient of determination if 70% or more than reliable .7 se

Coefficient of determination:

$R^2 = (cor)^2$

$= (0.75)^2 = 0.5625$

$= 56.25\%$

$R^2 = 0.99$

$r = 0.994....$

It means 56.25% of variation in the dependant variable is explained by its linear relation with independent variable

if slope value is -ive, relation is -ive

→ probability
→ Basic terms used in probability
→ Events and its types
(favourable outcomes)